

MathVerse AI

AI-Powered Gamified Mathematics Learning Platform

MathVerse AI transforms traditional mathematics practice into an interactive and personalized learning experience.

- Gamified math missions
- AI-powered tutor
- Student memory and personalization
- Eye-tracking based engagement analysis
- Adaptive difficulty using learning patterns
- Parent and teacher monitoring dashboard

Tagline: *“Learn. Battle. Adapt. Master.”*

1 Higher-order goal

This project aims to improve mathematics learning by reducing the gap between a student’s actual learning behavior and the support provided by conventional practice systems. While the topic is broader than a single mathematics curriculum, the immediate engineering objective is to translate personalized learning into a measurable, deployable software solution.

2 Time-to-value

- We will deliver meaningful increments quickly by prototyping the core student workflow and validating the learning experience within a short iteration window.
- We will prioritize fast feedback over perfection by implementing the mission flow, question engine, progress tracking, and core AI assistance early.
- Success is defined by what can be demonstrated and measured in the first iteration, then improved using student interaction data and feedback.

3 Problem Statement

Challenges in Traditional Mathematics Learning

- Mathematics practice can become repetitive and demotivating.
- The same difficulty is often provided to students with different abilities.
- Wrong answers do not always reveal the reason behind the mistake.
- Student attention and engagement are difficult to measure.
- Teachers and parents have limited visibility into how a student learns.
- Previous performance is rarely used effectively to personalize future practice.

Why this matters: A learning platform should respond not only to whether an answer is correct, but also to the student’s pace, repeated mistakes, hint usage, and interaction patterns.

4 Proposed Solution

Turning Mathematics into an Interactive Mission

Build a web-based MathVerse AI platform with the following components:

- **Student mission portal:** students select story-based mathematics missions and solve questions through an interactive interface.
- **Gamified question engine:** questions use health, timers, XP, rewards, objectives, and progression to create a mission-oriented experience.
- **AI tutor:** Gwen provides contextual hints, explanations, and mistake analysis using the student’s current question and learning history.
- **Adaptive learning engine:** question difficulty is adjusted using performance and behavioral signals.
- **Student memory:** recent attempts and long-term mastery information are stored for personalization.
- **Analytics dashboard:** students, teachers, and parents can view performance, mastery, trends, and engagement indicators.

4.1 Core workflow

1. Student opens the Hero Dashboard and selects a mission.
2. The system loads questions according to topic, current mastery, and mission difficulty.
3. Student solves questions while the system records correctness, response time, hint usage, and interaction signals.
4. Gwen provides guidance when the student requests help or when the system detects a useful intervention point.
5. The system updates mastery and behavioral memory after each attempt.
6. The adaptive engine selects an appropriate next difficulty or remediation activity.
7. Mission results are reflected in the student and parent/teacher dashboards.

4.2 Operational constraints

- Keep the solution usable on low-to-mid range devices through responsive web design.
- Treat eye-tracking as an additional behavioral signal rather than a standalone diagnosis.
- Use syllabus-aligned retrieval for AI explanations so generated guidance remains relevant to the curriculum.
- Store only the student information required for personalization, progress tracking, and analytics.

5 Solution Approach

Engineering Focus

The system focuses on scalable learning workflow, personalization, analytics, and measurable adaptation rather than claiming research-level novelty.

5.1 Student memory and personalization

Short-term memory: current mission, recent mistakes, current difficulty, recent hints, and recent response behavior.

Long-term memory: topic mastery, strengths, weaknesses, repeated mistakes, learning patterns, and historical performance.

5.2 RAG-based AI tutor

The AI tutor uses Retrieval-Augmented Generation to retrieve relevant syllabus-based material before producing an explanation. The retrieved context can include topic definitions, worked examples, formulas, and approved learning material. This separates curriculum retrieval from response generation and helps keep explanations aligned with the selected topic.

5.3 LSTM-based time analysis

An LSTM model analyzes sequences of learning behavior rather than treating each question independently. A representative sequence is:

Question → Response Time → Correctness → Hint Usage → Next Question

The model can learn temporal patterns such as increasing response time, repeated hesitation, changing hint usage, or improvement across successive questions. These signals can complement correctness when deciding whether to continue, remediate, or increase difficulty.

5.4 Adaptive difficulty

The adaptive engine combines performance and behavioral signals to select the next activity. For example, sustained high accuracy with efficient response behavior can move the learner from Easy → Medium → Hard. Repeated errors, high response time, or repeated hint usage can trigger Difficulty ↓ → Remediation → Practice → Difficulty ↑.

5.5 Eye-tracking and engagement analysis

Eye-tracking can provide non-invasive visual attention signals such as time spent on a question, attention shifts, re-reading behavior, and areas receiving more visual attention. These measurements are treated as complementary evidence alongside answer accuracy, response time, and interaction behavior.

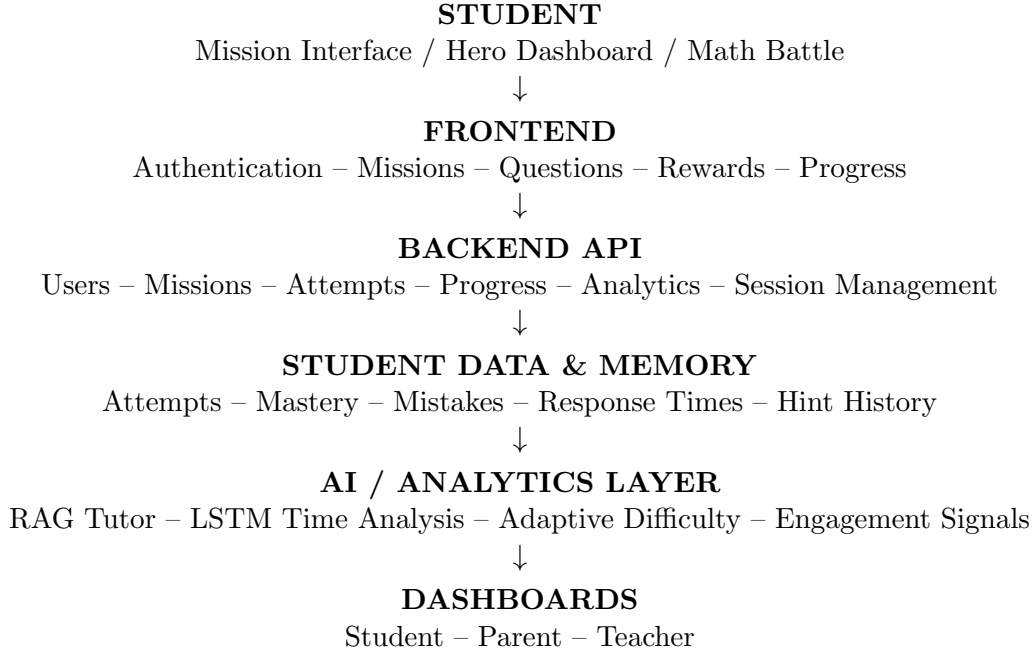
5.6 Evaluation criteria

- Accuracy improvement across repeated practice sessions.
- Reduction in repeated mistakes for identified weak topics.
- Change in response time across successive questions.
- Hint usage and subsequent correctness.
- Engagement indicators such as session completion and interaction consistency.

6 System Architecture

MathVerse AI follows a layered architecture in which the student-facing application communicates with backend services, persistent student memory, analytics modules, and AI components.

6.1 Architecture flow



6.2 Data flow

The student submits an answer → the backend records the attempt → the analytics layer processes correctness, response time, hint usage, and available behavioral signals → student memory is updated → the adaptive engine selects the next activity → the dashboard reflects the updated learning state.

6.3 Main components

Frontend: responsive student interface for missions, questions, battles, rewards, and progress.

Backend: authentication, mission management, question delivery, attempt logging, progress updates, and analytics APIs.

Database: users, questions, curriculum metadata, attempts, mastery records, hint history, mission state, and dashboard metrics.

AI layer: RAG-based tutor, mistake analysis, LSTM-based temporal analysis, and adaptive decision support.

Behavioral layer: eye-tracking and interaction signals used as complementary indicators of attention and difficulty.

6.4 Parent / Teacher Dashboard

- Overall performance
- Topic-wise mastery
- Weak areas
- Time spent learning
- Progress trends

- Engagement indicators
- Recommended interventions

7 Expected Impact

The system is designed to provide measurable benefits for each stakeholder in the learning process.

Students

- Higher engagement through mission-based practice.
- Personalized learning based on current mastery.
- Better conceptual understanding through contextual explanations.

Teachers

- Identify weak topics and recurring mistakes.
- Track individual progress and learning trends.
- Receive actionable insights for intervention.

Parents

- Monitor learning progress.
- Understand strengths and weaknesses.
- View learning patterns rather than only final scores.

8 Scalability

The platform should scale in both theoretical and practical terms by separating frontend and backend components, using indexed database queries and pagination for common dashboard operations, and keeping backend services stateless where possible.

Operational deployment can use common web hosting, managed databases, containerized or platform-hosted services, and CI/CD for repeatable staging and production releases.

9 Project Scope and Deliverables

Initial deliverable

- Student authentication and Hero Dashboard.
- Mission and question workflow.
- Attempt logging and progress tracking.
- Basic student memory and mastery calculation.
- Core Gwen AI assistance.
- Basic parent/teacher analytics dashboard.

Subsequent deliverables

- RAG-based syllabus retrieval.
- LSTM-based temporal analysis.
- Adaptive difficulty engine.

- Eye-tracking integration and engagement indicators.
- Advanced dashboards and personalized learning paths.

10 Risks and Mitigations

- **AI reliability:** ground explanations in syllabus-based retrieval and evaluate generated responses.
- **Privacy:** minimize collected student data and treat eye-tracking information carefully.
- **Model overinterpretation:** use behavioral signals as supporting evidence, not definitive judgments.
- **Evaluation ambiguity:** instrument attempts, timestamps, correctness, and difficulty transitions before testing.

11 Summary

MathVerse AI proposes a deployable web platform that combines gamification, AI tutoring, student memory, behavioral analysis, and adaptive learning. It focuses on measurable personalization: understanding what the student knows, identifying where the student struggles, and using learning behavior to determine what should happen next.

The final vision is simple: MathVerse AI does not just evaluate the answer — it understands the learner.